Submission Date : 11/3/2026

Day 1 – Designing RDBMS Tables Assignments

There are 10 different scenarios. You need to do the following tasks as a Database Modeler or Database Designer:

1) Create the **table designs** ensuring that all the columns are relevant in that table, as per the subject area of that table.
2) Consider appropriate data type for each column such as Text/String or Numeric or Date.
3) Show the relationships within the table appropriately.
4) Apart from Primary Key & Foreign Key, proper usage of Unique Key, Not Null, Check constraints and Composite Keys must be done.
5) Provide default data values for the columns wherever necessary.

Note → All above points should be shown in a **Tabular Format** for each scenario's each table.

Scenario 1: University Course Registration System

FacultyPhones

EnrollmentDate

CourseNames

StudentEmail

CourseCredits

FacultyNames

StudentAddress

DepartmentLocation

CourseIDs

StudentID

FacultySpecialization

StudentName

DepartmentName

CourseFee

FacultyID

DepartmentID

Semester

StudentMobile

CourseDuration

Scenario 2: Hospital Management System

DoctorRoomNumber

PatientEmail

TreatmentCost

InsuranceProvider

DoctorSpecialization

PatientMobile

DepartmentName

AppointmentDate

PatientName

PrescriptionDetails

DoctorPhone

DepartmentLocation

TreatmentName

PatientAddress

DoctorID

PatientID

DepartmentID

InsurancePolicyNumber

TreatmentDuration

DoctorName

Scenario 3: Banking & Loan Management System

BranchLocation

LoanType

CustomerEmail

LoanInterestRate

EmployeePhone

CustomerName

AccountType

BranchManagerName

CustomerID

LoanAmount

AccountBalance

BranchID

LoanStartDate

EmployeeID

CustomerAddress

AccountNumber

BranchName

EmployeeName

AccountOpenDate

LoanID

CustomerMobile

Scenario 4 → E-Commerce Order System

ProductPrice

OrderDate

CustomerEmail

ProductName

OrderID

CustomerAddress

ProductCategory

CustomerMobile

Quantity

PaymentMethod

CategoryDescription

CustomerName

OrderStatus

ProductID

CustomerID

ShippingCity

ShippingPincode

Scenario 5 → Hotel Booking Management System

RoomType

BookingDate

HotelCity

GuestEmail

RoomRate

GuestMobile

HotelName

RoomNumber

PaymentMethod

HotelAddress

GuestName

RoomCapacity

CheckOutDate

GuestAddress

BookingID

HotelID

CheckInDate

RoomStatus

GuestID

Scenario 6 → Retail Store Inventory Management System

SupplierEmail

ProductCategory

PurchaseDate

SupplierAddress

WarehouseLocation

ProductName

QuantityPurchased

CategoryDescription

ProductPrice

SupplierPhone

ProductID

WarehouseName

PurchaseID

ProductStock

SupplierName

SupplierID

WarehouseID

ReorderLevel

Scenario 7 → HR Payroll Management System

EmployeePhone

PayrollMonth

DepartmentLocation

EmployeeName

BasicSalary

TaxAmount

DepartmentName

EmployeeEmail

NetSalary

PayrollYear

AllowanceAmount

EmployeeAddress

Designation

EmployeeID

DepartmentID

DeductionAmount

PayDate

BankAccountNumber

Scenario 8 → Car Rental Management System

CarModel

RentalStartDate

CustomerPhone

CarCategory

RentalAmount

BranchLocation

CustomerName

ReturnDate

CarRegistrationNumber

InsuranceProvider

BranchName

CustomerEmail

CarDailyRate

RentalID

CustomerAddress

BranchID

InsurancePolicyNumber

CustomerID

CarStatus

Scenario 9 → Telecom Billing Management System

PlanName

CustomerEmail

CallDuration

PlanMonthlyCharge

CustomerAddress

TowerLocation

BillMonth

CustomerPhone

BillAmount

CustomerID

PlanType

CallDate

TowerID

BillYear

CustomerName

SIMNumber

PlanID

CallCharge

PaymentStatus

Scenario 10 → University Examination Result System

StudentName

ExamMonth

SubjectName

DepartmentName

StudentEmail

SubjectCredits

ExamYear

Grade

DepartmentLocation

ExamType

StudentMobile

FacultyName

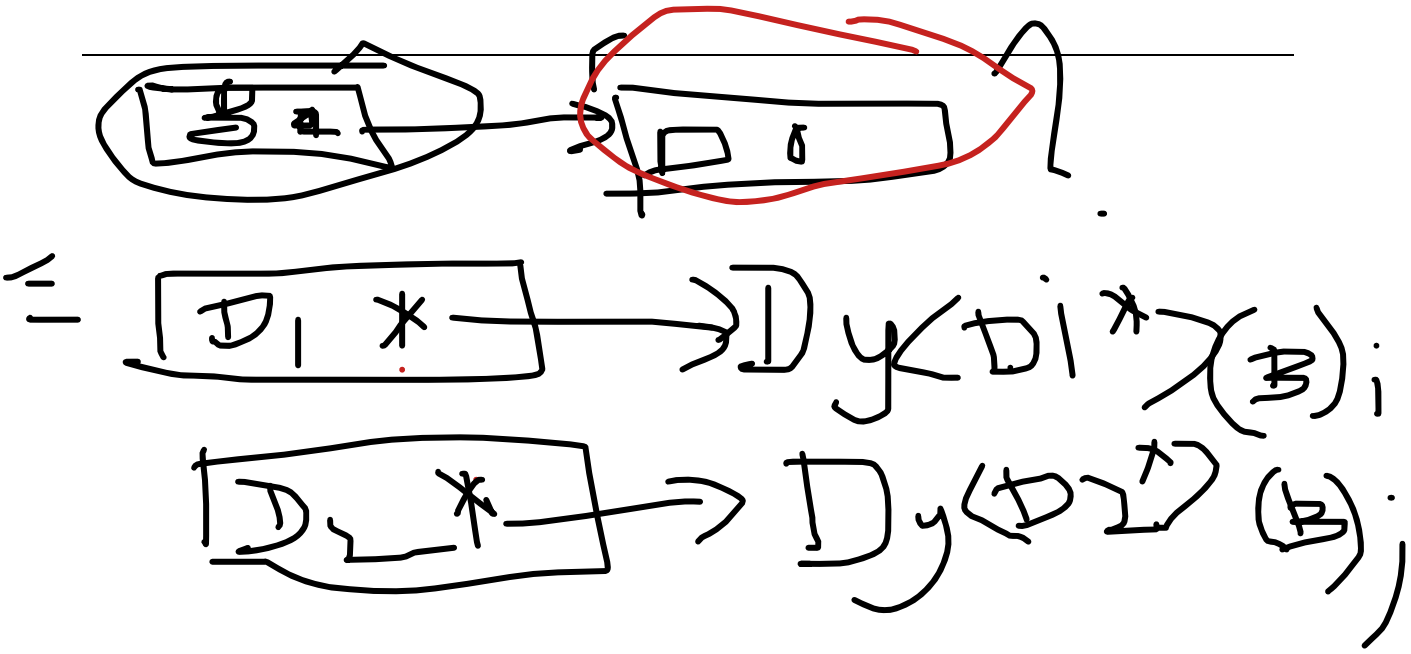
MarksObtained

StudentID

FacultyPhone

SubjectID

DepartmentID

ResultStatus

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